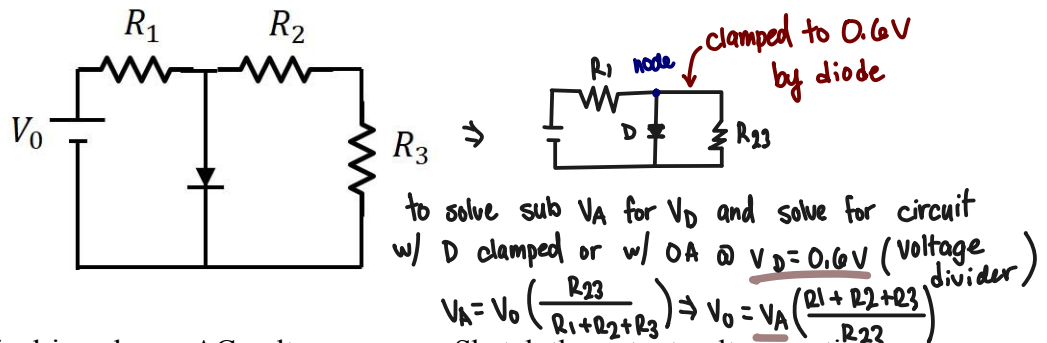


PHYS0525 Analog and Digital Electronics – Spring 2026

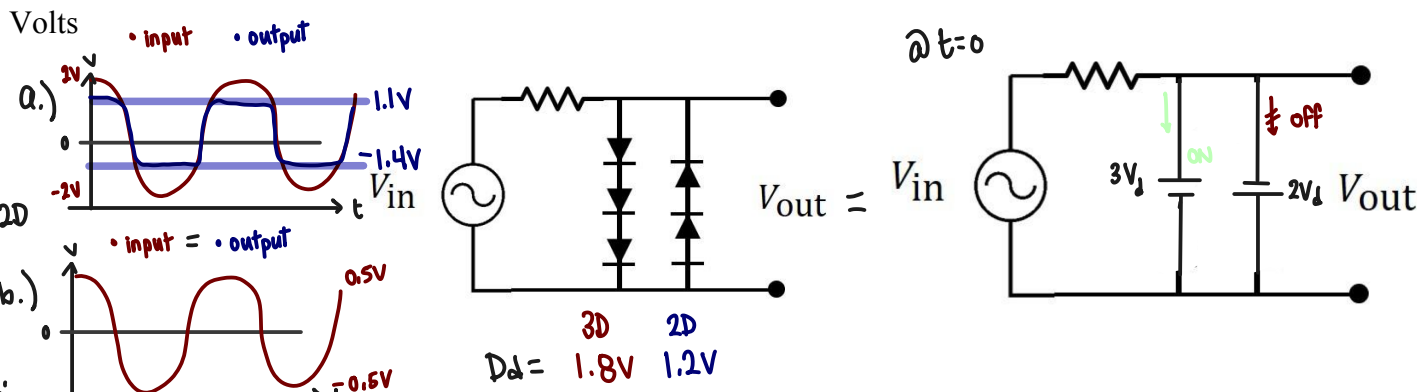
Assignment 5

Due Wed, Feb 25, 2026

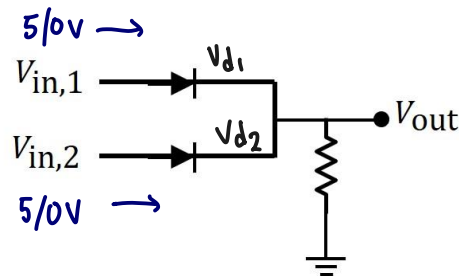
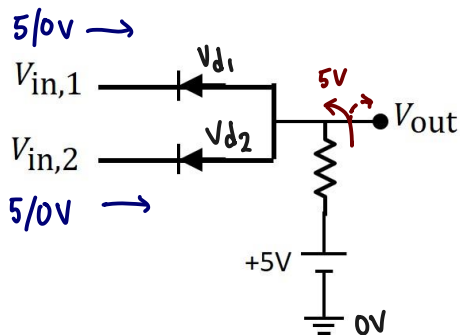
- 1) For the circuit shown below, what is maximum voltage V_0 that will result in zero current through the diode. Assume the diode is an ideal diode with a diode drop of 0.6 V.



- 2) The circuit shown below is driven by an AC voltage source. Sketch the output voltage vs time for a few cycles of the wave when (a) $V_{in}(t) = 4 \cos(\omega t)$ Volts and (b) $V_{in}(t) = 0.5 \cos(\omega t)$ Volts



- 3) Two circuits are shown below that take two inputs each. Assuming the inputs are restricted to have discrete values of either 0 V or 5 V, determine the output voltage for each circuit for all combinations of input voltages. These are rudimentary logic circuits known as an AND gate and an OR gate. Which circuit would you call AND, and which circuit would you call OR?



diodes act like one way switches, only one will win out!